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1 Abstract

Electric vehicle charging infrastructure continues to advance as global transportation shifts toward lower-emission technologies. This work investigates the Swiss Rectifier as an ac–dc conversion stage suitable for modern fast-charging applications. Multiple configurations are examined, including diode-based and thyristor-controlled implementations, each evaluated under both highly inductive and purely resistive loading conditions. The study highlights the rectifier’s ability to maintain proper conduction sequences, achieve full-wave operation, and adapt its behavior according to the selected switching devices and load characteristics. Overall, the analysis provides a clear comparison of Swiss Rectifier operating modes and demonstrates its suitability for reliable and efficient EV charging applications.

2 Introduction

Global transportation contributes nearly one-third of greenhouse gas emissions, driving rapid growth in electric vehicles as a cleaner alternative [1], [2]. EV adoption has surged past ten million units globally despite economic challenges, supported by declining lithium-ion battery prices and expanding fast-charging infrastructure [3]. Yet widespread deployment still faces barriers such as battery cost, limited driving range, long charging times, and grid impacts, including harmonics and unmanaged load increases. Recent reviews highlight advances in EV technologies and charging infrastructure, but detailed comparisons of power converter topologies and control strategies—especially for fast and multiport chargers—remain limited [4], [5]. This work examines the Swiss Rectifier ac–dc converter architecture for modern fast-charging systems.

3 Analysis and Case Studies

3.1 Swiss Rectifier

Swiss Rectifiers (SRs) [6] improve efficiency, reduce noise, and simplify control compared to six-switch TPBRs [7]. Interleaved designs offer lower ripple and high reliability, with reported efficiencies up to 99.3% at 8 kW [8]. Variants such as multilevel and full-bridge SRs enable higher power and better grid performance. Their main limitation is unidirectional power flow, with bidirectional operation achievable only through added complexity.

3.2 Diodes with Highly Inductive Load

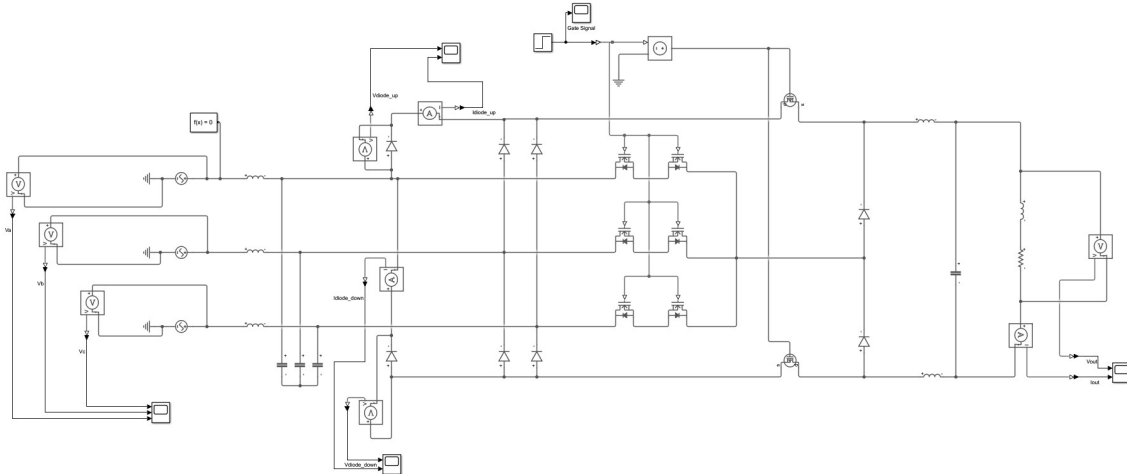


Figure 1: Swiss Rectifier with Diodes Implementation

Figure 1 shows the implemented Swiss Rectifier. The model uses a three-phase AC input with series inductors ($L = 1 \text{ mH}$) and parallel capacitors ($C = 1 \mu\text{F}$) for current conditioning, feeding a core structure composed of upper and lower diode paths and an eight-switch active network that enables simplified control, power-factor correction, and efficient boost regulation. For simplicity, the switches are switched on during the entire operation. Interleaved boost legs reduce current ripple, lower filter requirements, and improve EMI performance, while the DC-link capacitors ($C = 1 \mu\text{F}$) and output filter capacitors deliver a stable, high-quality DC output for fast-charging applications. The rectifying circuit is loaded with a highly inductive load (HIL) ($L = 1 \text{ H}$, $R = 1 \Omega$). Note that the diodes and switches exhibit certain non-ideal characteristics.

Figure 2a shows the balanced three-phase, shifted by 120° , input voltages applied to the Swiss Rectifier. Each phase— V_a , V_b , and V_c —has a peak value of 338.4 V , indicating $V_{rms} = 240 \text{ V}$. The rectified output is shown in Figure 2b. The output voltage increases from zero and settles at approximately $200\text{--}220 \text{ V}$, showing a switching-induced ripple of about $20\text{--}30 \text{ V}$. Likewise, the output current rises smoothly from zero to nearly $100\text{--}120 \text{ A}$, reflecting normal load-charging behavior. This steady convergence of both V_{out} and I_{out} confirms stable DC-link regulation and effective power transfer through the rectifier.

In the positive diode branch shown in Figure 2c, current conducts during the positive half-cycles, producing sharp current pulses and a reverse-biased voltage when not conducting. Conversely, the negative diode branch depicted in Figure 2d carries current during the negative half-cycles with similar pulse amplitude and switching-induced voltage ripple. The complementary conduction of both branches ensures that each half-cycle of the three-phase input contributes to the DC output, confirming proper full-wave rectification and correct interaction with the interleaved boost and switching network.

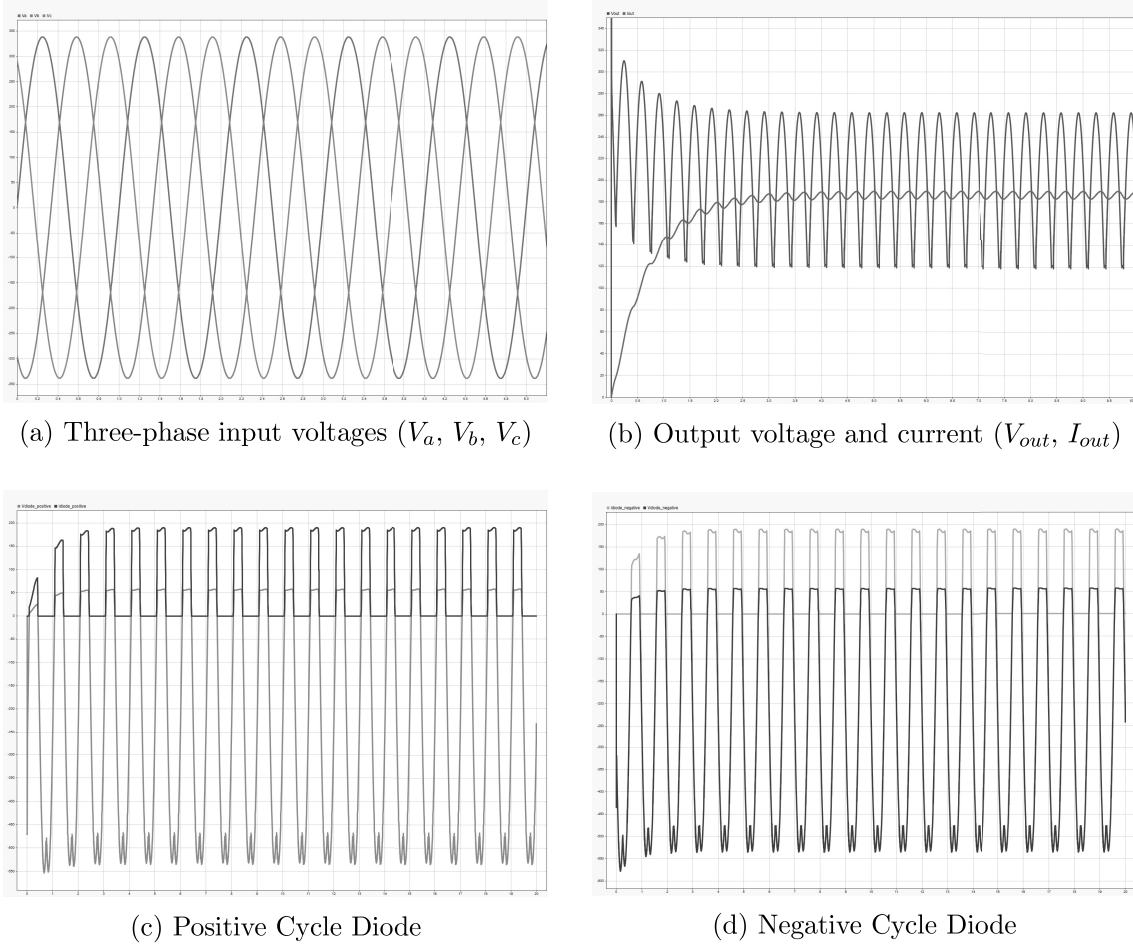


Figure 2: Input, output, and device waveforms for HIL (Diode Implementation)

3.3 Diodes with Resistive Load

In this section, the highly inductive load used in Section 3.2 is replaced with a purely resistive load of $R = 10 \, \Omega$, while maintaining the same three-phase input conditions as in the HIL setup. Figure 3 presents the corresponding waveforms of the Swiss Rectifier under this resistive loading. The output voltage varies between approximately 160 and 320 V, exhibiting noticeably larger ripple compared to the inductive case due to the absence of energy storage in the load. The load current follows the rectified voltage waveform directly, oscillating between 16 and 32 A. Both the positive-cycle and negative-cycle diodes demonstrate correct rectification behaviour, with a peak reverse voltage of approximately $V_{PRV} = -580 \, \text{V}$.

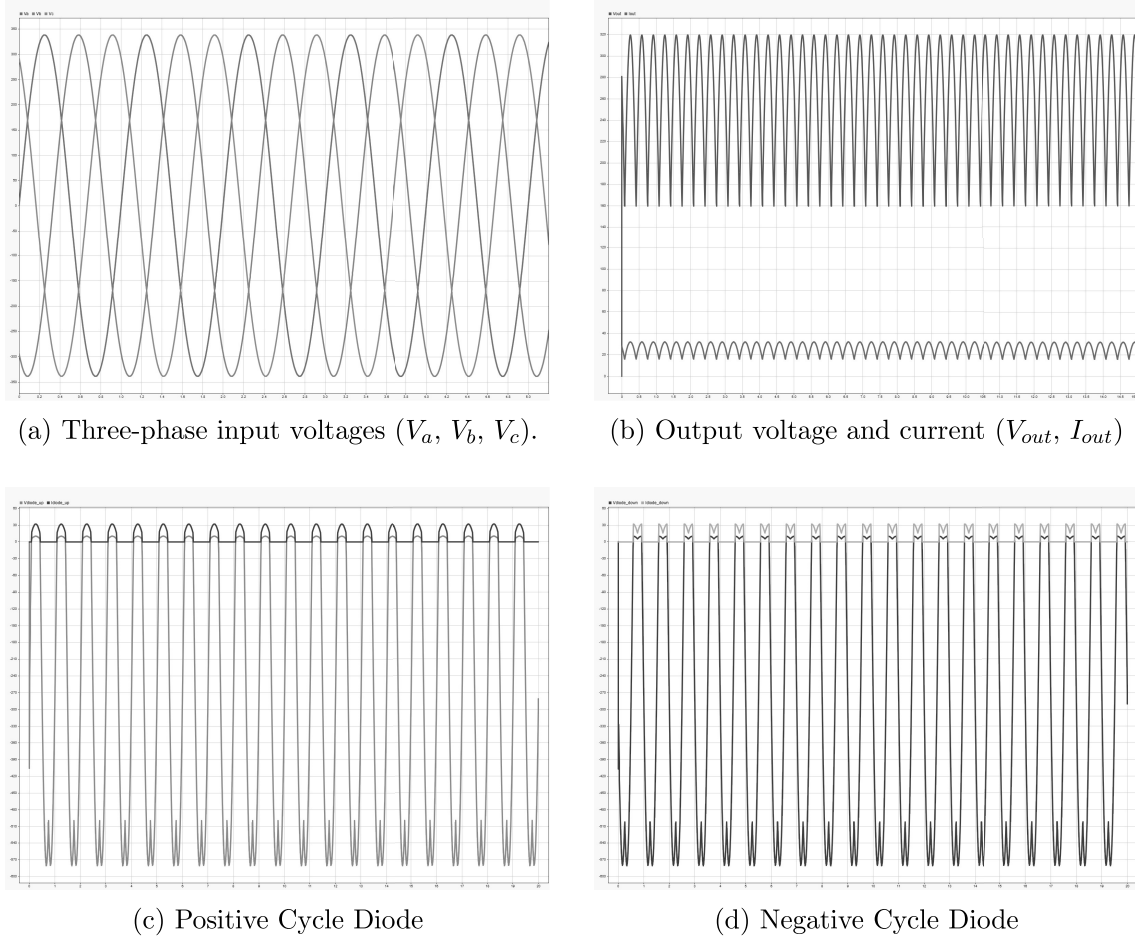


Figure 3: Input, output, and device-level waveforms for R-Load (Diode Implementation)

3.4 Thyristors with Highly Inductive Load

The positive-cycle diodes are removed from Figure 4 and replaced with thyristors, while the negative-cycle diodes remain in place to maintain the required freewheeling path. The same highly inductive load used in Section 3.2 is applied here ($L = 1$ H, $R = 1$ Ω), together with the same three-phase input source. A firing angle of $\alpha = 30^\circ$ (measured after the phase reference) is used for the thyristor triggering.

The firing pulses in Figure 6a occur once per cycle with a delay corresponding to the firing angle $\alpha = 30^\circ$ from the phase reference angle. The rectified output shown in Figure 6b exhibits a DC link voltage that rises from zero and stabilizes with peak values of approximately 260–270 V, accompanied by a switching ripple of 20–30 V. The output current ramps up smoothly and settles around 190 A. Device-level behavior confirms correct operation: the positive-cycle thyristor (Figure 5c) conducts at the appropriate firing instants, and the negative-cycle diode (Figure 5d) conducts on complementary half-cycles, experiencing reverse voltages in the range of -450 to -550 V.

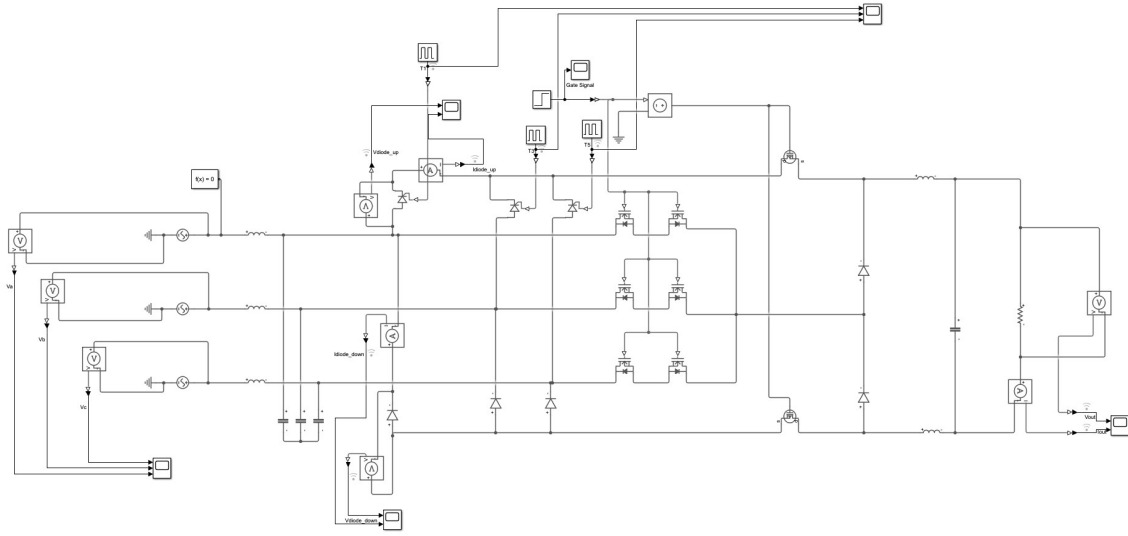
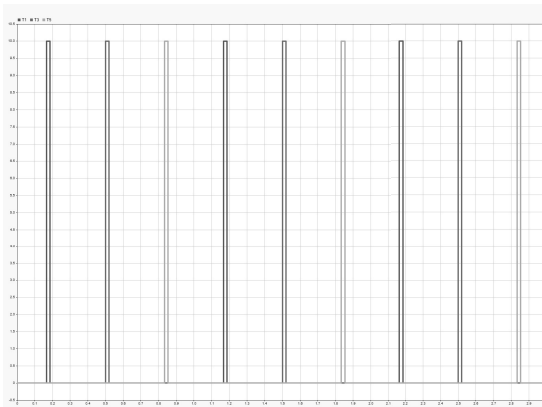
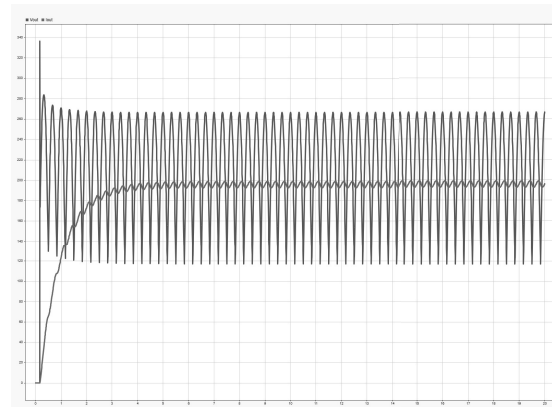


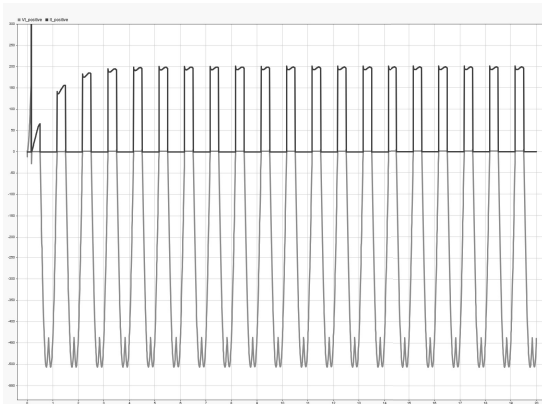
Figure 4: Swiss Rectifier with Thyristor Implementation



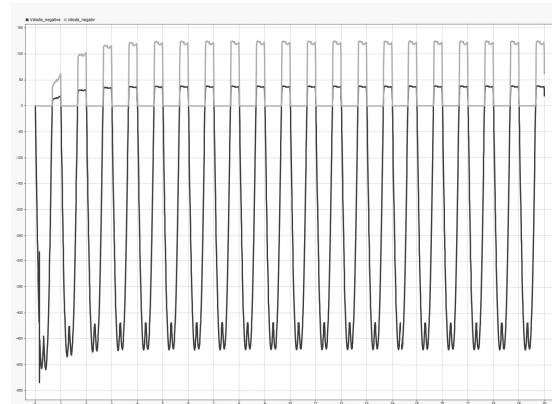
(a) Thyristor firing pulses (T1, T3, T5).



(b) Output voltage and current (V_{out} , I_{out}).



(c) Positive Cycle Thyristor

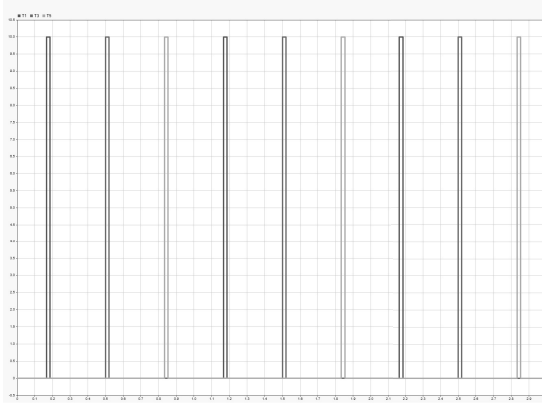


(d) Negative Cycle Thyristor

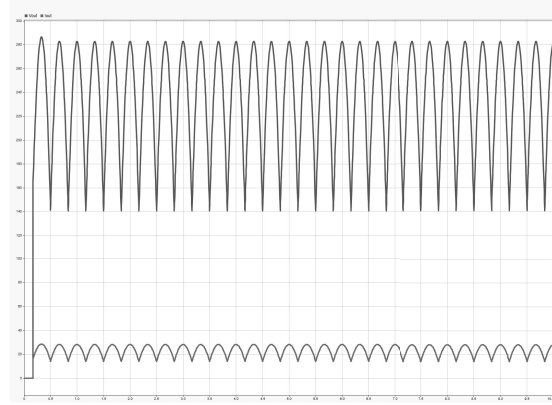
Figure 5: Firing, output, and device waveforms for HIL (Thyristor Implementation)

3.5 Thyristors with Resistive Load

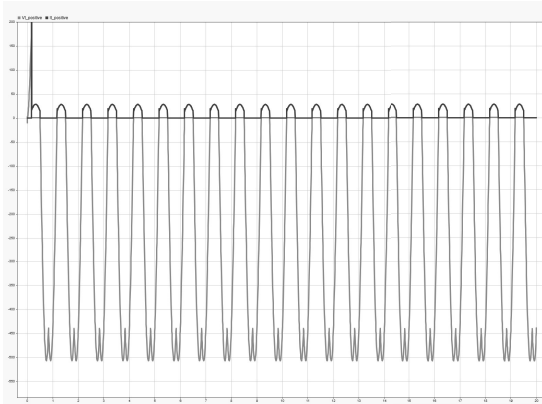
The HIL in Section 3.4 is replaced with the same resistive load in Section 3.3 ($R = 10\ \Omega$), using the same firing angle $\alpha = 30^\circ$ and same input voltage sources. Figure 6 shows the thyristor-based Swiss Rectifier feeding the resistive load; the gate pulses fire once per cycle with the expected 30° delay (Figure 6a). The output voltage (Figure 6b) is about 150–280 V and the load current is 20–28 A. Device plots (Figures 6c–6d) confirm correct conduction and blocking.



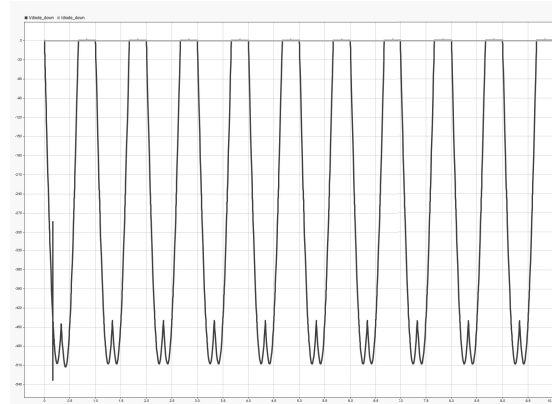
(a) Thyristor firing pulses (T1, T3, T5).



(b) Output voltage and current (V_{out} , I_{out}).



(c) Positive Cycle Thyristor



(d) Negative Cycle Diode

Figure 6: Firing, output, and device waveforms for R-Load (Thyristor Implementation)

4 Conclusion

The simulation results confirm the correct operation of the Swiss Rectifier under both diode-based and thyristor-controlled configurations for highly inductive and resistive loading conditions. For the diode implementation with a highly inductive load ($L = 1$ H, $R = 1$ Ω), the output voltage stabilizes around 200–220 V with a ripple of 20–30 V, while the load current reaches 100–120 A. Under a resistive load of $R = 10$ Ω , the diode rectifier produces a higher-ripple output of 160–320 V and a corresponding current of 16–32 A. For the thyristor implementation with a firing angle of $\alpha = 30^\circ$, the highly inductive load yields a rectified output of 260–270 V and a steady current near 190 A, whereas the resistive load results in an output voltage of 150–280 V and a load current of 20–28 A. These findings demonstrate stable rectification, proper device conduction, and effective power transfer across all tested operating conditions.

References

- [1] U. S. E. P. Agency, *Inventory of u.s. greenhouse gas emissions and sinks*, Online, Available: <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks>, Apr. 2021.
- [2] J. A. Krosnick and B. MacInnis, “Climate insights 2020,” *Surveying American public*, 2020.
- [3] I. E. Agency. “Global ev outlook.” <https://iea.blob.core.windows.net/assets/ed5f4484-f556-4110-8c5c-4ede8bcba637/GlobalEVOutlook2021.pdf>.
- [4] M. Adhikari, L. P. Ghimire, Y. Kim, P. Aryal, and S. B. Khadka, “Identification and analysis of barriers against electric vehicle use,” *Sustainability*, vol. 12, no. 12, p. 4850, 2020.
- [5] G Krishna, “Understanding and identifying barriers to electric vehicle adoption through thematic analysis,” *Transportation Research Interdisciplinary Perspectives*, vol. 10, p. 100364, 2021.
- [6] T. B. Soeiro, T. Friedli, and J. W. Kolar, “Swiss rectifier—a novel three-phase buck-type pfc topology for electric vehicle battery charging,” in *2012 Twenty-Seventh Annual IEEE Applied Power Electronics Conference and Exposition (APEC)*, IEEE, 2012, pp. 2617–2624.
- [7] M. Safayatullah, M. T. Elrais, S. Ghosh, R. Rezaii, and I. Batarseh, “A comprehensive review of power converter topologies and control methods for electric vehicle fast charging applications,” *IEEE Access*, vol. 10, pp. 40753–40793, 2022.
- [8] L. Schrittwieser, M. Leibl, M. Haider, F. Thöny, J. W. Kolar, and T. B. Soeiro, “99.3% efficient three-phase buck-type all-sic swiss rectifier for dc distribution systems,” *IEEE Transactions on Power Electronics*, vol. 34, no. 1, pp. 126–140, 2018.